Assignment No. 3

BT401–Genetic Resources and Conservation

Total Marks: 10

Due date: 04.08.21

Assignment questions

1. **What is cryopreservation? Which tissue can be preserved using this technique also explain merits and demerits of cryopreservation (5)?**

Cryopreservation is a technique in which biological material such as DNA, RNA and protein are storage at a very low temperature to prevent it to damage environment. Genetic resources of animals and plants can be also being preserved by the cryopreservation. Genetic resources may be in the form of cells, tissue and DNA. The DNA is extract from the cells and store at a very low temperature -80ºC. Cell also extracts from the plants and animals store by cryopreservation because all the enzymes in the cell cannot damage the cell because of enzyme are not activate at this very low temperature. The DNA Material may be stored because individuals shared specific genes and phenotypes that may be of value or have potential value for researchers or breeders. Cryopreservation serves as a way to preserve germplasms, which is particularly beneficial for threatened breeds.

Tissues which are preserved through cryopreservation:

1. Uniform tissues (skin tissue, cartilage tissue).

2. Initial cell population (sperm and egg).

3. Precursor cells (bone marrow tissue).

Merits of cryopreservation:

1. It helps in the preservation of biological materials like egg and sperms.

2. By using cryopreservation biological materials can be preserved for a long time.

3. Different types of cells or tissues are preserved like Sperm, gametes, embryos, tissues, bone marrow.

4. It is usually helping to study the adapting nature of plants and animals under the low temperature.

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Demerits of cryopreservation:

1. During the process of freezing the biological material may be becomes freeze and ice formation.

2. Due to migration of the water in the cell. It can dehydrate the cell and cell damage directly.

3. Technical difficulty in the process of organ cryopreservation.

2. **Mass extinction or biotic crisis is widespread and rapid decreases in biodiversity on Earth, Discuss probable causes. (2)**

Mass extinction or biotic crisis causes the change in the climate and most the biodiversity cannot be existing at this.

Causes:

1. Habitat destruction

2. Climate change

3. Overexploitation

3. **Differentiate between Ex-Situ and In-situ Conservation (3).**

Ex-Situ Conservation	In-situ Conservation
It is usually in artificial environment.	It is usually in Natural environment.
It is not cheap and used for commercial uses.	It is usually cheap and convenient.
ZOO and aquarium	Natural parks and wildlife

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